BRIGHTLEARN TUTORIALS

DATA SCIENCE

SQL EXERCISE 2 : AGGREGATE FUNCTIONS & GROUPING

Database : employees_db

1. COUNT () FUNCTION

SQL query to find the total number of employees

syntax :

```
SELECT COUNT(*) AS total_employees
FROM employees;
```

Output :

total_employees
10

2. SUM () FUNCTION

SQL query to find total salary paid to all employees in IT department

Syntax :

```
SELECT sum(salary) AS total_IT_salary
FROM employees
WHERE department = 'IT';
```

Output :

total_IT_salary
220000

3. AVG Function

SQL query to calculate the average salary of employees in HR department

Syntax :

```
SELECT AVG (salary) AS avg_HR_salary
FROM employees
WHERE department = 'HR';
```

Output :

avg_HR_salary
49500

4. MIN () and MAX functions

SQL query to find highest and lowest paid in the company

Syntax :

```
SELECT
    MIN (salary) AS lowest_paid,
    MAX (salary) AS highest_paid
FROM employees;
```

Output :

lowest_paid	highest_paid
48000	62000

5. GROUP BY Statement

SQL query to group employees by department and total salary paid in each department

Syntax :

```
SELECT department,
SUM (salary) AS total_salary
FROM employees
GROUP BY department;
```

Output :

department	total_salary
IT	220000
HR	99000
Finance	119000
Marketing	105000

6. GROUP BY and COUNT ( )

SQL query to count employees working in each city

Syntax :

```
SELECT city,
  COUNT (*) AS employee_count
FROM employees
GROUP BY city;
```

Output :

city	employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

SQL query to group employees by department, calculate average salary per department and order results in descending order

Syntax :

```
SELECT department,
AVG (salary) AS avg_salary
```

```
FROM employees
GROUP BY department
ORDER BY avg_salary DESC;
```

Output :

department	avg_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

8. HAVING Clause

SQL query to find department where total salary paid exceeds 100 000

Syntax :

```
SELECT department,
SUM (salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100000;
```

Output :

department	total_salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING and ORDER BY

SQL query to list cities where more than one employee works, ordered by the number of employees in descending order

Syntax :

```
SELECT city,
  COUNT (*) AS employee_count
FROM employees
GROUP BY city
HAVING COUNT (*) >1
ORDER BY employee_count DESC;
```

Output :

city	employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10. Combining Aggregate Functions

SQL query to find department with the highest average salary

Syntax :

```
SELECT department,
AVG (salary) As avg_salary_highest
FROM employees
GROUP BY department
ORDER BY avg_salary_highest DESC
LIMIT 1;
```

Output :

department	avg_salary_highest
Finance	59500